

For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps a) Getting the dataset. b) Importing libraries. c) Importing datasets. d) Finding Missing Data. e) Encoding Categorical Data

```
#GETTING DATA SET
import pandas as pd
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Toyota.csv to Toyota (1).csv

```
#IMPORTING DATASET
import pandas as pd
import io
toyotadf=pd.read_csv(io.BytesIO(uploaded['Toyota.csv']),index_col=0)
toyotadf
```

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Wei
0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1
1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1
2	13950	24.0	41711	Diesel	90	NaN	0	2000	3	1
3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1
4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1
...	...	...	...	...	...	...	...	...	...	...
1431	7500	NaN	20544	Petrol	86	1.0	0	1300	3	1
1432	10845	72.0	??	Petrol	86	0.0	0	1300	3	1
1433	8500	NaN	17016	Petrol	86	0.0	0	1300	3	1
1434	7250	70.0	??	NaN	86	1.0	0	1300	3	1
1435	6950	76.0	1	Petrol	110	0.0	0	1600	5	1

1436 rows × 10 columns

step1: information

```
toyotadf.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1436 entries, 0 to 1435
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Price       1436 non-null   int64
1   Age         1336 non-null   float64
2   KM          1436 non-null   object
3   FuelType    1336 non-null   object
4   HP          1436 non-null   object
5   MetColor    1286 non-null   float64
6   Automatic   1436 non-null   int64
7   CC          1436 non-null   int64
8   Doors       1436 non-null   object
9   Weight      1436 non-null   int64
dtypes: float64(2), int64(4), object(4)
memory usage: 123.4+ KB
```

step 2: checking on missing values

```
toyotadf.isnull()
```

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	W
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	True	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...	
1431	False	True	False	False	False	False	False	False	False	
1432	False	False	False	False	False	False	False	False	False	
1433	False	True	False	False	False	False	False	False	False	
1434	False	False	False	True	False	False	False	False	False	
1435	False	False	False	False	False	False	False	False	False	

1436 rows × 10 columns

```
toyotadf.isnull().sum()
```

```
Price      0
Age        100
KM          0
FuelType   100
HP          0
MetColor   150
Automatic   0
```

```
CC          0
Doors       0
Weight     1000
dtype: int64
```

```
import numpy as np
np.unique(toyotadf['KM'])
```

```
array(['1', '10000', '100123', ..., '99865', '99971', '??'], dtype=object)
```

```
import numpy as np
np.unique(toyotadf['HP'])
```

```
array(['107', '110', '116', '192', '69', '71', '72', '73', '86', '90',
      '97', '98', '????'], dtype=object)
```

```
import numpy as np
np.unique(toyotadf['Automatic'])
```

```
array([0, 1])
```

```
import numpy as np
np.unique(toyotadf['Doors'])
```

```
array(['2', '3', '4', '5', 'five', 'four', 'three'], dtype=object)
```

step3: replacing missing values in different formats to NAN

```
import pandas as pd
import io
toyotadf=pd.read_csv(io.BytesIO(uploaded['Toyota.csv']),index_col=0,na_value
toyotadf
```

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors
0	13500	23.0	46986.0	Diesel	90.0	1.0	0	2000	three
1	13750	23.0	72937.0	Diesel	90.0	1.0	0	2000	3
2	13950	24.0	41711.0	Diesel	90.0	NaN	0	2000	3
3	14950	26.0	48000.0	Diesel	90.0	0.0	0	2000	3
4	13750	30.0	38500.0	Diesel	90.0	0.0	0	2000	3
...	...	...	...	...	...	...	...	...	...
1431	7500	NaN	20544.0	Petrol	86.0	1.0	0	1300	3
1432	10845	72.0	NaN	Petrol	86.0	0.0	0	1300	3
1433	8500	NaN	17016.0	Petrol	86.0	0.0	0	1300	3
1434	7250	70.0	NaN	NaN	86.0	1.0	0	1300	3
1435	6950	76.0	1.0	Petrol	110.0	0.0	0	1600	5

1436 rows × 10 columns

```
toyotadf.isnull().sum()
```

```
Price      0
Age       100
KM         15
FuelType   100
HP          6
MetColor   150
Automatic   0
CC          0
Doors       0
Weight     0
dtype: int64
```

```
toyotadf['Doors'].replace('three',3,inplace=True)
toyotadf['Doors'].replace('four',4,inplace=True)
toyotadf['Doors'].replace('five',5,inplace=True)
toyotadf
```

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors
0	13500	23.0	46986.0	Diesel	90.0	1.0	0	2000	3
1	13750	23.0	72937.0	Diesel	90.0	1.0	0	2000	3
2	13950	24.0	41711.0	Diesel	90.0	NaN	0	2000	3
3	14950	26.0	48000.0	Diesel	90.0	0.0	0	2000	3
4	13750	30.0	38500.0	Diesel	90.0	0.0	0	2000	3
...	...	...	...	...	...	...	...	...	...
1431	7500	NaN	20544.0	Petrol	86.0	1.0	0	1300	3
1432	10845	72.0	NaN	Petrol	86.0	0.0	0	1300	3
1433	8500	NaN	17016.0	Petrol	86.0	0.0	0	1300	3
1434	7250	70.0	NaN	NaN	86.0	1.0	0	1300	3
1435	6950	76.0	1.0	Petrol	110.0	0.0	0	1600	5

1436 rows × 10 columns

```
toyotadf['Doors'].unique()
```

```
array([3, '3', '5', '4', 4, 5, '2'], dtype=object)
```

```
toyotadf['Doors']=toyotadf['Doors'].astype('int')  
toyotadf['Doors'].unique()
```

```
array([3, 5, 4, 2])
```

```
toyotadf
```

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors
0	13500	23.0	46986.0	Diesel	90.0	1.0	0	2000	3
1	13750	23.0	72937.0	Diesel	90.0	1.0	0	2000	3
2	13950	24.0	41711.0	Diesel	90.0	NaN	0	2000	3
3	14950	26.0	48000.0	Diesel	90.0	0.0	0	2000	3
4	13750	30.0	38500.0	Diesel	90.0	0.0	0	2000	3
...	...	...	...	...	...	...	...	...	...
1431	7500	NaN	20544.0	Petrol	86.0	1.0	0	1300	3
1432	10845	72.0	NaN	Petrol	86.0	0.0	0	1300	3
1433	8500	NaN	17016.0	Petrol	86.0	0.0	0	1300	3
1434	7250	70.0	NaN	NaN	86.0	1.0	0	1300	3
1435	6950	76.0	1.0	Petrol	110.0	0.0	0	1600	5

1436 rows × 10 columns

```
toyotadf.isnull().sum()
```

```
Price      0
Age       100
KM         15
FuelType   100
HP          6
MetColor   150
Automatic   0
CC          0
Doors       0
Weight      0
dtype: int64
```

```
toyotadf.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1436 entries, 0 to 1435
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Price       1436 non-null   int64
1   Age         1336 non-null   float64
2   KM          1421 non-null   float64
3   FuelType    1336 non-null   object
4   HP          1430 non-null   float64
5   MetColor    1286 non-null   float64
6   Automatic    1436 non-null   int64
7   CC          1436 non-null   int64
8   Doors       1436 non-null   int64
9   Weight      1436 non-null   int64
dtypes: float64(4), int64(5), object(1)
```

```
memory usage: 123.4+ KB
```

to check total number of missing values in the dataset

```
toyotadf.isnull().sum().sum()
```

```
371
```

```
toyotadf.describe()
```

	Price	Age	KM	HP	MetColor	Au
count	1436.000000	1336.000000	1421.000000	1430.000000	1286.000000	1436
mean	10730.824513	55.672156	68647.239972	101.478322	0.674961	0
std	3626.964585	18.589804	37333.023589	14.768255	0.468572	0
min	4350.000000	1.000000	1.000000	69.000000	0.000000	0
25%	8450.000000	43.000000	43210.000000	90.000000	0.000000	0
50%	9900.000000	60.000000	63634.000000	110.000000	1.000000	0
75%	11950.000000	70.000000	87000.000000	110.000000	1.000000	0
max	32500.000000	80.000000	243000.000000	192.000000	1.000000	1

for replacing the missing values in the dataset we use fillna...

we are replacing the missing values with mean values

```
toyotadf['Age'].fillna(toyotadf['Age'].mean(), inplace=True)
```

```
toyotadf.isnull().sum()
```

```
Price      0
Age        0
KM         15
FuelType   100
HP          6
MetColor   150
Automatic   0
CC          0
Doors       0
Weight     0
dtype: int64
```

```
toyotadf['KM'].fillna(toyotadf['KM'].median(), inplace=True)
```

```
toyotadf.isnull().sum()
```

```
Price      0
Age        0
KM         0
FuelType   100
HP         6
MetColor   150
Automatic   0
CC         0
Doors      0
Weight     0
dtype: int64
```

```
toyotadf['FuelType'].value_counts()
```

```
Petrol    1177
Diesel     144
CNG        15
Name: FuelType, dtype: int64
```

```
toyotadf['FuelType'].fillna(toyotadf['FuelType'].value_counts().index[0], in
```

```
toyotadf['MetColor'].value_counts()
```

```
1.0    868
0.0    418
Name: MetColor, dtype: int64
```

```
toyotadf['MetColor'].mode()
```

```
0    1.0
dtype: float64
```

```
toyotadf['MetColor'].fillna(toyotadf['MetColor'].mode().index[0], inplace=Tr
```

```
toyotadf['HP'].fillna(toyotadf['HP'].mean(), inplace=True)
```

```
toyotadf.isnull().sum()
```

```
Price      0
Age        0
KM         0
FuelType    0
HP         0
MetColor    0
Automatic   0
CC         0
Doors      0
Weight     0
dtype: int64
```

1.uploaded the data set

2.get the data set

3.info()

4.check for missing values using isnull().sum()

5.check for uniques values in all columns using np.unique

6.identify the different unique values and replace them with na_values

7.use fillna to replace missing values with mean, median and mode

toyotadf

	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Wei
0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1
1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1
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3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1
4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1
...	...	...	...	...	...	...	...	...	...	...
1431	7500	NaN	20544	Petrol	86	1.0	0	1300	3	1
1432	10845	72.0	??	Petrol	86	0.0	0	1300	3	1
1433	8500	NaN	17016	Petrol	86	0.0	0	1300	3	1
1434	7250	70.0	??	NaN	86	1.0	0	1300	3	1
1435	6950	76.0	1	Petrol	110	0.0	0	1600	5	1

1436 rows × 10 columns

```
toyotadfnew=pd.get_dummies(toyotadf,columns=['FuelType'])
toyotadfnew
```

Price	Age	KM	HP	MetColor	Automatic	CC	Doors	Weight	FuelT
-------	-----	----	----	----------	-----------	----	-------	--------	-------